# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** magnet | Removes iron. |
| 2 | **(d)** unsaturated | More solute can dissolve. |
| 3 | **(d)** to avoid spurting | Splashing. |
| 4 | **(a)** residue | It stays on paper. |
| 5 | **(b)** crystallises out | Solubility falls. |
| 6 | **(c)** to hold the paper well | Good contact. |
| 7 | **(b)** salt solution | Salt dissolved. |
| 8 | **(a)** solution became saturated | Extra salt remains. |
| 9 | **(c)** pure water is collected | Distillation. |
| 10 | **(a)** evaporation | Heat. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Sand does not dissolve (1); Sedimentation and filtration are suitable (1). | 1+1 |
| 12 | Both dissolve (1); They pass through together (1). | 1+1 |
| 13 | It remains behind (1); The clear liquid is the filtrate (1). | 1+1 |
| 14 | Use a magnet to remove iron (1); Sugar remains (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Handpick stones (1); Use magnet for nails (1); Sieve out sand (1). | 1+1+1 |
| 16 | Large surface area (1); Sun heats quickly (1); Faster evaporation (1). | 1+1+1 |
| 17 | Chalk particles are tiny (1); They pass through the holes along with the water (1); Sedimentation or filtration is better (1). | 1+1+1 |
| | **OR:** Heat a few drops on a glass slide (1); Water evaporates (1); White solid remains (1). | 1+1+1 |
| 18 | Pure substance has one kind of matter (1); Mixture has more (1); Examples: gold vs sea water (1). | 1+1+1 |
| | **OR:** If solid is insoluble: filtration (1); If dissolved: evaporation (1); Check solubility (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Magnet removes iron (1); Add water (1); Sawdust floats and is skimmed (1); Filter sand (1); Evaporate to get sugar (1). | 1+1+1+1+1 |
| | **OR:** Winnowing and sieving (1); Straining (1); Evaporation (1); Churning (1); Basic ideas same (1). | 1+1+1+1+1 |
| 20 | Take two glasses with equal water (1); One cold, one warm (1); Add salt spoon by spoon (1); Count spoons dissolved (1); Warm water dissolves more (1). | 1+1+1+1+1 |
| | **OR:** Heat the solution (1); Steam to a cool vessel (1); Water collected (1); Solid remains (1); Distillation (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Choosing filtration for salt in water; dissolved salt passes through the filter, so **evaporation** is needed.
- Mixing **sedimentation** (settling) with **decantation** (pouring off the clear liquid).
- Mixing **threshing** (beating stalks to free grain) with winnowing (blowing away husk).
- Calling the liquid that passes through filter paper the residue; it is the **filtrate**.
- Using a magnet to separate sand or salt; only magnetic substances like iron are attracted.
- Believing that filtered or settled water is always safe to drink.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** evaporation of salt solution | Salt can be recovered. |
| 2 | **(a)** churning | Spins. |
| 3 | **(b)** decantation after two layers form | Immiscible liquids. |
| 4 | **(b)** sand and water | Insoluble solid. |
| 5 | **(b)** winnowing and handpicking | Combine. |
| 6 | **(a)** sedimentation | Gravity. |
| 7 | **(c)** sugar and water | Sugar dissolves. |
| 8 | **(c)** evaporate the water | Crystallisation. |
| 9 | **(a)** filtration | Strainer. |
| 10 | **(d)** magnet then dissolving and evaporating | Combination. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Size, weight, solubility differ (1); A suitable method uses one such property (1). | 1+1 |
| 12 | Use a sieve whose holes let sand pass but not the seeds (1); Seeds are larger than sand grains (1). | 1+1 |
| 13 | Add water; sugar dissolves (1); Filter and evaporate the filtrate (1). | 1+1 |
| 14 | They mix completely (1); They form a single layer, so nothing can be poured off (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | All have holes (1); Hole size differs (1); Used for different sizes (1). | 1+1+1 |
| 16 | Stones are heavier and settle (1); Decant dirty water (1); Rice is clean (1). | 1+1+1 |
| 17 | Threshing frees grain from stalk (1); Winnowing removes husk (1); Both done in succession (1). | 1+1+1 |
| | **OR:** Oil floats as a separate layer (1); Salt dissolves uniformly (1); Distillation or evaporation used for salt (1). | 1+1+1 |
| 18 | Layers of gravel, sand (1); Pour water (1); Dirt trapped (1). | 1+1+1 |
| | **OR:** Use a magnet (1); Nails stick to it (1); Beads remain (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Sea water let into pans (1); Sun evaporates water (1); Salt crystals appear (1); Collected and dried (1); Purified for use (1). | 1+1+1+1+1 |
| | **OR:** Solution: salt in water (1); The dissolved salt cannot be seen (1); Mixture of solids: sand and sugar (1); Parts of a solid mixture can be seen (1); Parts of a solution do not settle (1). | 1+1+1+1+1 |
| 20 | Settle (1); Alum (1); Cloth filter (1); Boil (1); Store safely (1). | 1+1+1+1+1 |
| | **OR:** Stir salt in water (1); Disappears (1); Stir sand (1); Settles (1); Salt solution clear, sand residue (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Choosing filtration for salt in water; dissolved salt passes through the filter, so **evaporation** is needed.
- Mixing **sedimentation** (settling) with **decantation** (pouring off the clear liquid).
- Mixing **threshing** (beating stalks to free grain) with winnowing (blowing away husk).
- Calling the liquid that passes through filter paper the residue; it is the **filtrate**.
- Using a magnet to separate sand or salt; only magnetic substances like iron are attracted.
- Believing that filtered or settled water is always safe to drink.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** carries away dirt | Removes dirt. |
| 2 | **(a)** sieving | Different sizes. |
| 3 | **(a)** winnowing | Light and heavy. |
| 4 | **(b)** particles settle | Sedimentation. |
| 5 | **(d)** settle faster | Alum coagulates particles. |
| 6 | **(b)** dissolving sugar and filtering | Solubility differs. |
| 7 | **(d)** sieving | Size. |
| 8 | **(b)** sedimentation, decantation, filtration, boiling | Standard. |
| 9 | **(d)** evaporation | Evaporation is for dissolved solids in liquids. |
| 10 | **(a)** to remove iron pieces | Safety. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Holds the filter paper (1); Directs the filtrate (1). | 1+1 |
| 12 | Particles clump together (1); They settle faster (1). | 1+1 |
| 13 | Wind blows away light husk (1); Heavy grain falls down (1). | 1+1 |
| 14 | Evaporate and condense the vapour (1); Collect the droplets (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Heat a little sea water in a dish (1); The water evaporates (1); White salt is left behind (1). | 1+1+1 |
| 16 | Boil the water (1); Steam goes to a cool tube (1); Condenses (1). | 1+1+1 |
| 17 | Faster dissolving (1); Brings fresh water to the solute (1); Solution forms sooner (1). | 1+1+1 |
| | **OR:** Tea: strainer (1); Flour: sieve (1); Rice and stones: handpicking (1). | 1+1+1 |
| 18 | Use a strainer (1); Water passes (1); Rice remains (1). | 1+1+1 |
| | **OR:** Holds filter paper (1); Guides the liquid (1); Prevents spilling (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Heat flask (1); Steam goes through tube (1); Cooled in water (1); Condensed (1); Collect pure water (1). | 1+1+1+1+1 |
| | **OR:** Stones harm teeth (1); Iron bits dangerous (1); Sieves and magnets used (1); Hygiene (1); Quality (1). | 1+1+1+1+1 |
| 20 | Winnowing: weight (1); Magnet: magnetic property (1); Sedimentation: density (1); Evaporation: water turns to vapour but the dissolved solid does not (1); Sieving: size (1). | 1+1+1+1+1 |
| | **OR:** Filter juice (1); Boil to evaporate water (1); Cool (1); Crystals form (1); Separate and dry (1). | 1+1+1+1+1 |

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